

Program 8: Create Kubernetes Pods using YAML Descriptors

Project Structure

program8

- |--nodejs-configmap.yaml
- |--nodejs-pod.yaml
- |--[server.js](#)

Create a folder(p8)

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ mkdir prg8
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ cd prg8
bash: cd: prg8: No such file or directory
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ cd prg8
```

Step 1: Write a [Node.js](#) app

nano [server.js](#)

```
GNU nano 7.2 server.js
const http = require("http");

const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2: Create a ConfigMap for the code

```
GNU nano 7.2 nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create the port yaml using [Node.js](#) as a base image

```
GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-app
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs-container
      image: node:18-alpine
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app/server.js
          subPath: server.js
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: Apply the YAML files

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/prg8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/prg8$ kubectl apply -f nodejs-pod.yaml
The Pod "nodejs-app" is invalid: spec.initContainers: Forbidden: pod updates may not add or remove containers
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/prg8$ kubectl delete pod nodejs-app
pod "nodejs-app" deleted from default namespace
```

Step 5: Check pod status

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/prg8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-app created
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/prg8$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
hello-cron-29425611-fl98m	0/1	Completed	0	2m24s
hello-cron-29425612-245dr	0/1	Completed	0	84s
hello-cron-29425613-6qdlr	0/1	Completed	0	24s
kubia	1/1	Running	1 (48m ago)	4d
mypod	0/1	Error	0	11d
nginx-deploy-55dd9db9b8-5f648	1/1	Running	2 (3d23h ago)	11d
nginx-deploy-55dd9db9b8-cxdz9	1/1	Running	2 (3d23h ago)	11d
nginx-deploy-55dd9db9b8-vbzt	1/1	Running	2 (3d23h ago)	11d
nginx-deployment-6f9664446b-2f95d	1/1	Running	0	42m
nginx-deployment-6f9664446b-f5fhz	1/1	Running	0	42m
nginx-pod	0/1	Completed	0	11d
nodejs-app	0/1	ContainerCreating	0	14s

Step 6: Access the [Node.js](#) app

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/prg8$ kubectl port-forward pod/nodejs-app 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

Step 7: Check the output in <http://localhost:3000>



